Faithful Diffusion-Based Image Restoration

Trading off perceptual quality and faithfulness in conditional diffusion super-resolution

Oct 12, 2026 · @Shlok

Summary and research question

This project measures how much generative strength a diffusion super-resolution model can use before it starts inventing detail, and whether a reconstruction constraint buys back faithfulness. The task is 4x single-image super-resolution on standard benchmarks, compared against bicubic upscaling and a CNN regression model.

Research question. As the generative strength of a conditional diffusion model for 4x super-resolution increases, how does the trade-off between perceptual quality and faithfulness to the original image change, and can an explicit reconstruction constraint shift that trade-off toward faithfulness?

Sub-questions

1. **Quality:** Do outputs become sharper and perceptually closer to the original (lower LPIPS) as generative strength increases?
2. **Faithfulness:** Does stronger generation also add invented content, seen as lower PSNR and SSIM, worse consistency with the low-resolution input, and more disagreement between random seeds?
3. **Balance:** Is there an operating point (a strength setting, or a constrained model) that improves perceptual quality over the CNN baseline while keeping fidelity loss within a fixed tolerance?

Motivation and background

A sharper image is not automatically a more accurate one, because super-resolution is ill-posed: one low-resolution image is consistent with many high-resolution images.

- **Regression models** (CNNs trained with L1 or L2 loss) predict roughly the average of all plausible answers. The result is accurate per pixel but blurry, since averaging washes out texture.
- **Generative models** (diffusion, GANs) sample one plausible answer. The result looks sharp, but some of that sharp detail may never have existed in the original.

Blau and Michaeli (2018) showed this is a fundamental limit, the perception-distortion trade-off: past a certain point, no method can look more realistic without moving further from the ground truth in pixel terms. So the useful question is not whether diffusion beats regression on every metric. It is where a model sits on that curve and how precisely we can control its position.

Invented detail is harmless in casual photo enhancement but harmful wherever a restored image is treated as evidence or measurement: medical and satellite imagery, text and license plates, forensic footage, and robot perception. Typical failures are fur texture that does not match the real animal, letters that change identity, and windows or bricks added to a facade.

Most diffusion super-resolution papers report one operating point per model. The contribution here is a controlled sweep of a single model's generative strength, with fidelity, perceptual quality, and consistency measured at every point, so the trade-off is visible as a curve rather than two numbers.

Hypotheses

Each hypothesis maps to one sub-question and is stated against generative strength s, a number from 0 (no generation, output equals bicubic) to 1 (full generation from pure noise), defined under *Controlling generative strength*.

ID	Sub-question	Hypothesis	Rejected if
H1	Quality	LPIPS falls as s increases, with diminishing returns at high s	Spearman correlation between s and LPIPS is not significantly negative
H2	Faithfulness	PSNR, SSIM, and low-resolution consistency fall, and cross-seed variance rises, as s increases	Any of these trends is flat or reversed with 95% confidence
H3	Balance	The reconstruction-constrained model reaches the same LPIPS as the unconstrained model at higher PSNR	Its LPIPS vs PSNR curve does not lie above the unconstrained curve on held-out data

Problem setup and data

The input is a 4x bicubic-downsampled image y, the target is the original high-resolution image x, and every model outputs an estimate of x at full resolution.

```
y = (x \ast k)\downarrow_4 \qquad \hat{x} = f(y)
```

- **Primary degradation:** bicubic downsampling by 4 (k is the bicubic kernel). This matches the standard benchmark protocol, so baseline numbers are comparable to published ones.
- **Stretch degradation:** Gaussian blur (sigma 1.0 and 2.0) before downsampling, covering the blurry-input case in the original idea.
- **Training crops:** random 128x128 high-resolution patches (32x32 low-resolution), with flips and 90-degree rotations.
- **Evaluation:** full test images, tiled with overlap if the diffusion model runs out of memory.

Dataset	Images	Use	Why it is included
DIV2K train	800	Training	Standard high-resolution training set for super-resolution
DIV2K validation	100	Model selection, choosing the operating point	Held out from training and from final test numbers
Set5, Set14	5, 14	Test	Standard small benchmarks, for comparison with published results
BSD100	100	Test	Natural textures (fur, foliage) where invented detail looks plausible
Urban100	100	Test	Regular structures (windows, grids, text) where invented detail is easy to spot

Models compared

Four models span the range from no learning, to learned averaging, to learned sampling, to sampling with a faithfulness constraint.

#	Model	Role	Training objective	Inference	Expected behavior
1	Bicubic upscaling	Reference floor	None	One interpolation	Blurry, invents nothing
2	EDSR-baseline CNN (16 residual blocks, 64 channels)	Regression baseline	L1 loss to the high-resolution target	One forward pass	Highest PSNR and SSIM, smooth textures
3	Conditional diffusion (SR3-style U-Net)	Main model	Noise-prediction loss, conditioned on the bicubic-upsampled input	DDIM sampler, strength s	Sharp, may invent detail as s rises
4	Reconstruction-constrained diffusion (stretch)	Faithfulness variant	Model 3 loss plus a reconstruction term	Same as model 3	Tested for fewer invented details at similar sharpness

Model 2. Use the official EDSR-baseline x4 weights, which are trained on DIV2K like the other models. Confirm its Set5 PSNR matches the paper before using it as the baseline.

Model 3. The bicubic-upsampled input is concatenated to the noisy image as extra input channels, following SR3. The network learns to predict the noise added at step t:

```
\mathcal{L}_{\text{diff}} = \mathbb{E}_{x, \epsilon, t}\left[\lVert \epsilon
- \epsilon_\theta(x_t, t, \tilde{y}) \rVert_2^2\right], \qquad x_t =
\sqrt{\bar\alpha_t}\, x + \sqrt{1-\bar\alpha_t}\, \epsilon
```

Here y-tilde is the bicubic upsample of y, T = 1000 training steps, and sampling uses 50 DDIM steps.

Model 4. Same network and training budget as model 3. At each training step, the model's implied clean image is recovered from its noise prediction and pulled toward the ground truth:

```
\hat{x}_0 = \frac{x_t - \sqrt{1-\bar\alpha_t}\,\epsilon_\theta}
{\sqrt{\bar\alpha_t}}, \qquad \mathcal{L} = \mathcal{L}_{\text{diff}} +
\lambda\, \bar\alpha_t \lVert \hat{x}_0 - x \rVert_1
```

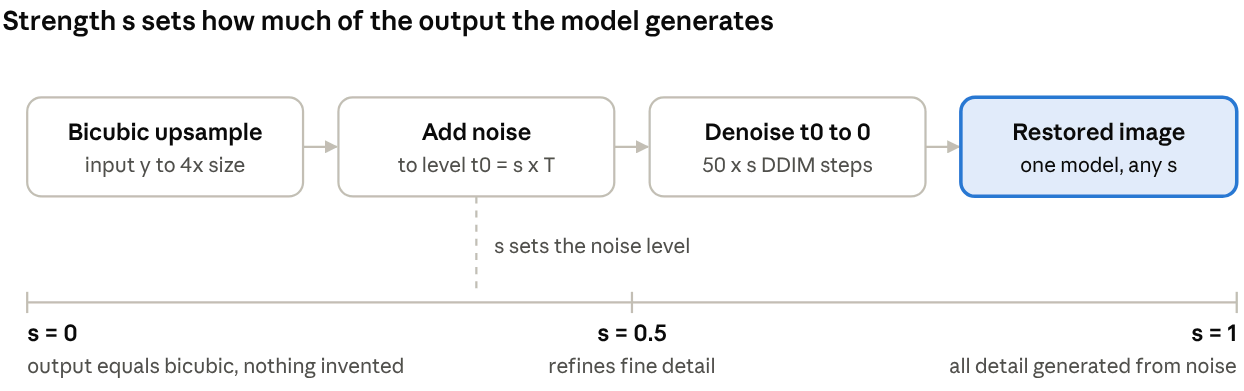
The weight alpha-bar-t down-weights very noisy steps, where the clean-image estimate is necessarily blurry. Lambda takes values 0.1 and 1.0. If there is no time to retrain, a fallback that needs no training is to project each sampling step's clean-image estimate so that its downsampled version equals y (back-projection, as in DDNM).

Controlling generative strength

Generative strength s is the fraction of the diffusion process the model runs, starting from a noised bicubic upsample instead of pure noise (the SDEdit approach). One trained model covers the whole range, so differences between settings come only from s.

```
t_0 = \lfloor s \cdot T \rfloor, \qquad x_{t_0} =
\sqrt{\bar\alpha_{t_0}}\,\tilde{y} + \sqrt{1-\bar\alpha_{t_0}}\,\epsilon,
\qquad \text{then denoise from } t_0 \text{ to } 0
```

- **s = 0:** no noise is added and no denoising happens, so the output is the bicubic upsample.
- **Small s:** a little noise is added, so the model can only refine fine detail; most content is inherited from the input.
- **s = 1:** sampling starts from pure noise, so all detail is generated, guided only by the conditioning input.
- **Sweep grid:** s in {0.1, 0.2, 0.3, 0.5, 0.7, 1.0}. The DDIM step spacing is fixed at T/50, so lower s also means fewer steps and faster inference. The time cost of each setting is itself a result.
- **Secondary knob (optional):** at s = 1, the DDIM noise parameter eta in {0, 0.5, 1}, which moves sampling from deterministic to fully stochastic.



strength control · 4 steps, one knob

Only the noise level in step 2 changes between settings; the trained model and sampler stay fixed.

Classifier-free guidance was considered and not used as the main knob. Raising the guidance scale on the low-resolution input increases adherence to that input, so it does not map cleanly onto how much detail is generated. The start step does.

Evaluation metrics

The four planned metrics answer sub-questions 1 and 3, but on their own they cannot separate a blurry output from a wrong one, so two faithfulness measures are added for sub-question 2.

Metric	What it measures	Better	Sub-question	Protocol
PSNR (dB)	Pixel-level accuracy against the original	Higher	2, 3	Y channel of YCbCr, 4-pixel border cropped (standard for 4x)
SSIM	Similarity of local structure, contrast, brightness	Higher	2, 3	Same as PSNR
LPIPS	Perceptual distance to the original	Lower	1, 3	AlexNet backbone, RGB scaled to [-1, 1]
Inference time (s)	Cost to restore one image	Lower	3	Median of 20 runs after 3 warm-up runs, batch 1, 512x512 output, same GPU, all sampling steps included
Low-resolution consistency (dB), added	Whether the output, downsampled again, matches the input	Higher	2	PSNR between bicubic-downsampled output and y

Metric	What it measures	Better	Sub-question	Protocol
Cross-seed variation, added	How much of the output the input fails to determine	Lower	2	Per-pixel standard deviation over 5 seeds, averaged per image; also saved as a map

Why the two additions. A lower PSNR can mean a slightly misplaced real edge or a fully invented one; the metric cannot tell them apart. Low-resolution consistency catches content that contradicts the input. It cannot catch invented content that happens to downsample correctly, since many different details average to the same pixels. Cross-seed variation covers that case: if five seeds disagree in a region, the model is guessing there.

Caveat on LPIPS. LPIPS measures perceptual similarity to the original, not realism by itself. If a pure realism score is needed later, FID on 256x256 crops can be added.

Experiments

Five experiments, run in order; E1 to E3 are the core project and E4 and E5 are added if time permits.

1. **E1, baselines.** Run bicubic and EDSR on all four test sets and record every metric except cross-seed variation (both are deterministic).
   - Gate: EDSR's Set5 PSNR matches the published value before moving on.
2. **E2, strength sweep (H1, H2).** Run the diffusion model at each s in the grid, 5 seeds per image, on all test sets.
   - Output: each metric plotted against s, mean with 95% confidence interval.
3. **E3, perception-distortion frontier (H3, sub-question 3).** Plot LPIPS against PSNR for bicubic, EDSR, and every diffusion setting, one chart per test set, and mark the Pareto-optimal points.
   - Operating point: on DIV2K validation, choose the largest s whose mean PSNR is within 1 dB of EDSR (tolerance adjustable), then report that setting on the test sets.
4. **E4, reconstruction-constrained diffusion (stretch, H3).** Train model 4 with lambda = 0.1 and 1.0, repeat the E2 sweep, and overlay its curves on the E3 chart.

5. **E5, qualitative audit of invented detail (stretch).** Select 20 crops from Urban100 and BSD100 containing text, faces or fur, and repeating patterns.
    - Show the original, bicubic, EDSR, diffusion at s = 0.3, 0.7, 1.0, and the cross-seed variation map side by side.
    - Two people label each crop without knowing the method: faithful, invented texture, or changed structure.
    - This checks whether the automatic faithfulness metrics agree with human judgment.

Analysis plan and expected results

All metrics are computed per image, and every claim is backed by a per-image statistical test rather than a difference in means.

Statistics

- Means with 95% bootstrap confidence intervals (10,000 resamples over images).
- Method vs method: Wilcoxon signed-rank test on paired per-image differences, Holm correction for multiple comparisons.
- Trend with s (H1, H2): Spearman correlation between s and each metric, per test set.
- Seeds: average the 5 seeds per image before testing, so images, not samples, are the unit of analysis.

Expected results (based on the perception-distortion theory and published SR3 results; to be confirmed)

- EDSR has the best PSNR and SSIM of all models.
- Diffusion at s = 1 has the best LPIPS but the lowest PSNR among learned models.
- As s rises, LPIPS improves while PSNR, SSIM, and low-resolution consistency fall and cross-seed variation rises, with the largest effects on Urban100.
- If H3 holds, the constrained model′s frontier sits above the unconstrained one, meaning it buys back PSNR at a given LPIPS.

A flat or reversed trend is a valid result. For example, if PSNR stays flat while LPIPS improves, the trade-off is not binding in this range of s, which is itself worth reporting.

Scope, risks, and mitigations

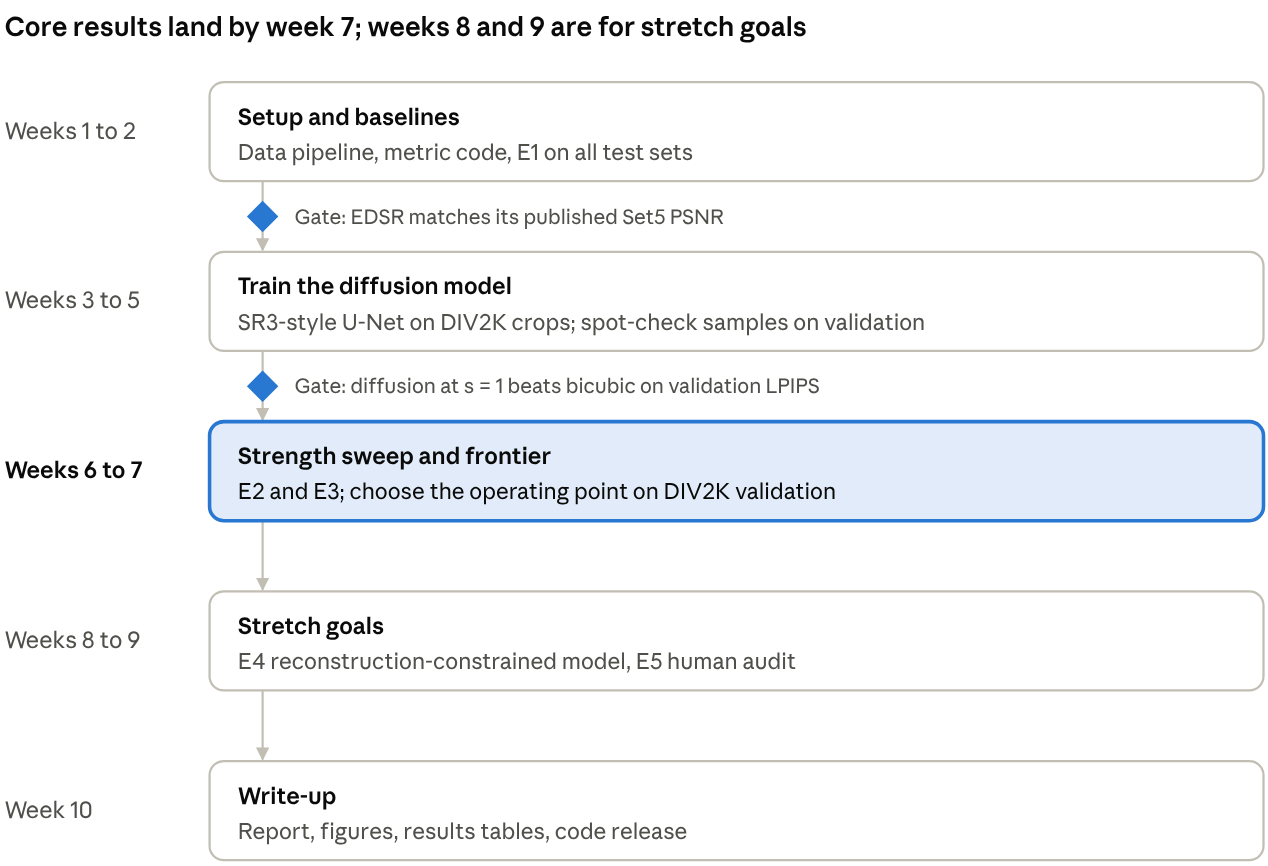
The biggest risk is diffusion training time, so every later experiment reuses one trained model and the stretch goals have no-training fallbacks.

Risk	Effect	Mitigation
Diffusion training takes too long	No model 3 in time for E2	Train on 128x128 crops with a mid-sized U-Net and a fixed step budget; fall back to fine-tuning a public SR3-style checkpoint
Full-image sampling runs out of GPU memory	Cannot evaluate on large test images	Tile at 256x256 with 32-pixel overlap and blend
Sweep is slow (6 strengths x 5 seeds x 319 images)	E2 does not finish	50 DDIM steps; low s runs fewer steps; cut seeds to 3 if needed
Automatic metrics disagree with what people see	Faithfulness conclusions are wrong	E5 human audit
No time to retrain model 4	H3 untested	Inference-time back-projection, which needs no training

Out of scope: real-world camera degradations, scale factors other than 4, and large pretrained latent-diffusion upscalers.

Timeline and deliverables

The plan assumes a 10-week quarter; shift the weeks if the deadline differs.



project plan · 5 phases, 2 gates, 10 weeks

The two gates stop the project from building on a broken baseline or an untrained model. If either slips, the stretch phase shrinks first.

Deliverables

- Code repository: data pipeline, all four models, metric scripts, sweep runner
- Results tables for every model and strength setting on all test sets
- Metric-vs-strength plots (E2) and perception-distortion frontier charts (E3)
- Qualitative figure with cross-seed variation maps (E5, if run)
- Final report following the structure of this document

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